ESOFT METRO CAMPUS

Shaping Lives, Creating Futures.

L3 – Diploma in Information Technology

Final Project

Campus One Digital Academy

[Student's Name]

[Registration No]

[Submission Date]

All rights reserved © ESOFT Metro Campus

Final Project – Assessment Sheet	
Programme	L3 - Diploma in Information Technology
Assessor/Lecturer	[Lecturer's name]
Unit	Final Project
Project Title	Campus One Digital Academy
Student's Name	[Your name]
Student's Registration No	[Your registration number]
Submission Date	[Submission date]
Marks Achieved (%)	
Assessor/Lecturer Feedback	

All rights reserved © ESOFT Metro Campus

1. Project Brief

[Paste the entire project brief (all nine pages) here, as required by the brief. Open the original brief .docx, press Ctrl+A then Ctrl+C, and paste it on this page. Delete this yellow note afterwards.]

All rights reserved © ESOFT Metro Campus

2. Acknowledgement

[Write this in your own words and thank the people who actually helped you, for example your lecturer and anyone who supported you. Delete this note and replace it with your own paragraphs.]

All rights reserved © ESOFT Metro Campus

3. Introduction

This project is a Windows Forms application developed in C# using Microsoft Visual Studio for Campus One Digital Academy. The system has two forms: a Login form and a Student Registration form. Only the Admin user can log in. After logging in, the user can register new students, search for a registered student by Reg No, update a record and delete a record. All student details are stored in a SQL Server database named Student.

The application uses the following tools and technologies:

- Microsoft Visual Studio (Windows Forms App, C#)
- Microsoft SQL Server Management Studio (database design)
- ADO.NET with Microsoft.Data.SqlClient (database connection)

4. Database Design

The database was created using SQL Server Management Studio. The database is named Student and it contains one table named Registration. The column regNo is the primary key. The data types follow the table structure given in the brief; the three phone number columns use BIGINT (a larger integer type) so that long phone numbers can be stored.

4.1 SQL Script

```
CREATE DATABASE Student;
GO
USE Student;
GO

CREATE TABLE Registration (
    regNo       INT PRIMARY KEY,
    firstName   VARCHAR(50),
    lastName    VARCHAR(50),
    dateOfBirth DATETIME,
    gender      VARCHAR(50),
    address     VARCHAR(50),
    email       VARCHAR(50),
    mobilePhone BIGINT,
    homePhone   BIGINT,
    parentName  VARCHAR(50),
    nic         VARCHAR(50),
    contactNo   BIGINT
);
```

4.2 Registration Table Structure

Field Name	Data Type	Notes
regNo	int	Primary key
firstName	varchar(50)	
lastName	varchar(50)	
dateOfBirth	datetime	
gender	varchar(50)	Male or Female

All rights reserved © ESOFT Metro Campus

Field Name	Data Type	Notes
address	varchar(50)	
email	varchar(50)	
mobilePhone	bigint	Integer type
homePhone	bigint	Integer type
parentName	varchar(50)	
nic	varchar(50)	
contactNo	bigint	Integer type

[Insert your own screenshot: SQL Server – Registration table design view (column names, types, primary key)]

Figure 4.1 – Registration table design

[Insert your own screenshot: SQL Server – Registration table, Edit Top 200 Rows with sample data]

Figure 4.2 – Registration table with sample records

All rights reserved © ESOFT Metro Campus

5. Form 1 – Login Form

The Login form is the first form shown when the system runs. The user enters the username and password. If the credentials are Admin and Campusone@123, the Login form is hidden and the Registration form opens. Otherwise an error message is displayed. The Clear button empties both text boxes and moves the cursor to the username box. The Exit button asks for confirmation before closing the system.

5.1 Design View

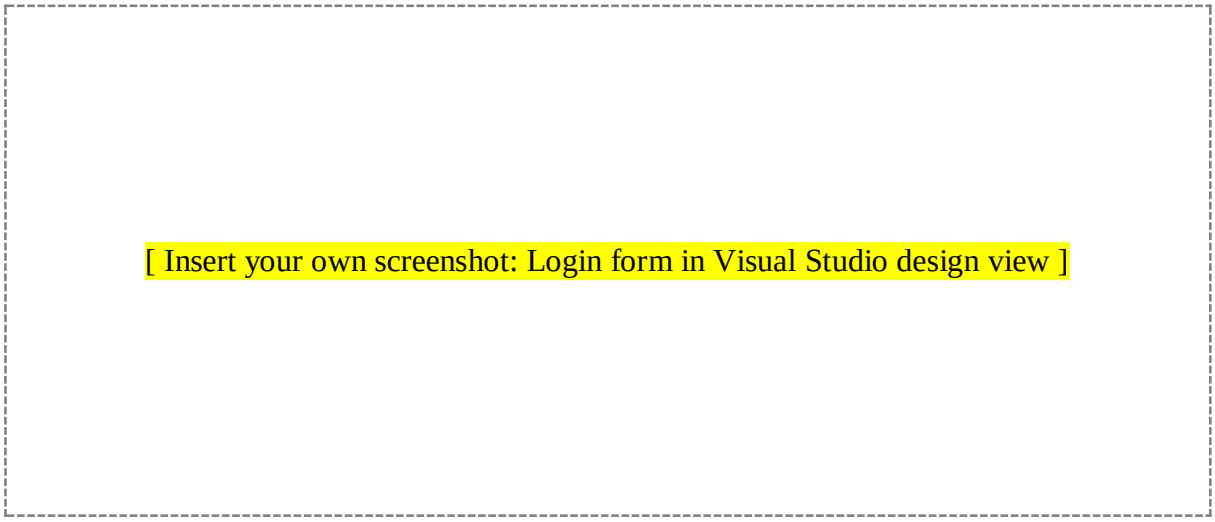


Figure 5.1 – Login form design

5.2 Controls and Properties

Replace the property values below with the values you actually set in your project.

Control	(Name)	Properties
Form	LoginForm	Text = Login - Campus One Digital Academy; StartPosition = CenterScreen; FormBorderStyle = FixedDialog; MaximizeBox = False
PictureBox	pictureBox1	Image = academy logo; SizeMode = Zoom
GroupBox	groupBox1	Text = Login; Font = Times New Roman, 12
Label	lblTitle	Text = Campus One Digital Academy; Font = Times New Roman, 20, Bold
Label	lblUsername	Text = Username; Font = Times New Roman, 12
Label	lblPassword	Text = Password; Font = Times New Roman, 12
TextBox	txtUsername	Font = Times New Roman, 12
TextBox	txtPassword	Font = Times New Roman, 12; UseSystemPasswordChar = True
Button	btnClear	Text = Clear; BackColor = [your colour]
Button	btnLogin	Text = Login; BackColor = [your colour]

All rights reserved © ESOFT Metro Campus

Control	(Name)	Properties
Button	btnExit	Text = Exit; BackColor = [your colour]

5.3 Code – LoginForm.cs

```
namespace CampusOneDigitalAcademy
{
    public partial class LoginForm : Form
    {
        public LoginForm()
        {
            InitializeComponent();
        }

        // Used by Clear button and when returning after Logout
        public void ResetForm()
        {
            txtUsername.Clear();
            txtPassword.Clear();
            txtUsername.Focus();
        }

        private void btnLogin_Click(object sender, EventArgs e)
        {
            if (txtUsername.Text == "Admin" && txtPassword.Text == "Campusone@123")
            {
                RegistrationForm reg = new RegistrationForm();
                reg.Show();
                this.Hide();
            }
            else
            {
                MessageBox.Show(
                    "Invalid Login credentials, please check " +
                    "Username and Password and try again",
                    "Invalid login Details",
                    MessageBoxButtons.OK, MessageBoxIcon.Error);
            }
        }

        private void btnClear_Click(object sender, EventArgs e)
        {
            ResetForm();
        }

        private void btnExit_Click(object sender, EventArgs e)
        {
            DialogResult result = MessageBox.Show(
                "Are sure, Do you really want to Exit...?", "Exit",
                MessageBoxButtons.YesNo, MessageBoxIcon.Question);

            if (result == DialogResult.Yes)
            {
                Application.Exit();
            }
        }
    }
}
```

5.4 Code – Program.cs

All rights reserved © ESOFT Metro Campus

Program.cs makes the Login form the first form to open.

```
namespace CampusOneDigitalAcademy
{
    internal static class Program
    {
        [STAThread]
        static void Main()
        {
            ApplicationConfiguration.Initialize();
            Application.Run(new LoginForm());   // Login form opens first
        }
    }
}
```

All rights reserved © ESOFT Metro Campus

6. Form 2 – Student Registration Form

The Registration form is used to manage student records. The Register button adds a new record to the Registration table. The Reg No drop-down list shows all registered numbers; selecting one fills in all the other fields from the database (search). The Update button saves changes to the selected record, the Delete button removes the selected record after confirmation, and the Clear button empties the form. The Logout link returns to the Login form and the Exit link closes the system after confirmation.

6.1 Design View

[Insert your own screenshot: Registration form in Visual Studio design view]

Figure 6.1 – Registration form design

6.2 Controls and Properties

Replace the property values below with the values you actually set in your project.

Control	(Name)	Properties
Form	RegistrationForm	Text = Student Registration - Campus One Digital Academy; StartPosition = CenterScreen
Label	lblTitle	Text = Campus One Digital Academy; Font = Times New Roman, 18, Bold
LinkLabel	lnkLogout	Text = Logout
LinkLabel	lnkExit	Text = Exit
GroupBox	grbStudent	Text = Student Registration
GroupBox	grbBasic	Text = Basic Details
GroupBox	grbContact	Text = Contact Details
GroupBox	grbParent	Text = Parent Details

All rights reserved © ESOFT Metro Campus

Control	(Name)	Properties
Label	lblRegNo	Text = Reg No
ComboBox	cmbRegNo	DropDownStyle = DropDown
Label	lblFirstName	Text = First Name
TextBox	txtFirstName	
Label	lblLastName	Text = Last Name
TextBox	txtLastName	
Label	lblDOB	Text = Date of Birth
DateTimePicker	dtpDOB	Format = Short
Label	lblGender	Text = Gender
RadioButton	rbMale	Text = Male
RadioButton	rbFemale	Text = Female
Label	lblAddress	Text = Address
TextBox	txtAddress	Multiline = True
Label	lblEmail	Text = Email
TextBox	txtEmail	
Label	lblMobile	Text = Mobile Phone
TextBox	txtMobile	MaxLength = 10
Label	lblHome	Text = Home Phone
TextBox	txtHome	MaxLength = 10
Label	lblParentName	Text = Parent Name
TextBox	txtParentName	
Label	lblNIC	Text = NIC
TextBox	txtNIC	MaxLength = 12
Label	lblContactNo	Text = Contact No
TextBox	txtContactNo	MaxLength = 10
Button	btnRegister	Text = Register
Button	btnUpdate	Text = Update
Button	btnClear	Text = Clear
Button	btnDelete	Text = Delete

6.3 Code – RegistrationForm.cs

Before running, replace YOUR_SERVER_NAME in the connection string with the server name shown in SQL Server Management Studio.

All rights reserved © ESOFT Metro Campus

```
using Microsoft.Data.SqlClient;

namespace CampusOneDigitalAcademy
{
    public partial class RegistrationForm : Form
    {
        // Replace YOUR_SERVER_NAME with the server name shown in SSMS
        private const string ConnStr =
            "Server=YOUR_SERVER_NAME;Database=Student;" +
            "Trusted_Connection=True;TrustServerCertificate=True;";

        private bool loggingOut = false;

        public RegistrationForm()
        {
            InitializeComponent();
        }

        private void RegistrationForm_Load(object sender, EventArgs e)
        {
            LoadRegNumbers();
        }

        // Closing this form with the X button closes the whole system
        private void RegistrationForm_FormClosed(
            object sender, FormClosedEventArgs e)
        {
            if (!loggingOut)
            {
                Application.Exit();
            }
        }

        // ---------- helper methods ----------
        private void LoadRegNumbers()
        {
            cmbRegNo.Items.Clear();
            using (SqlConnection con = new SqlConnection(ConnStr))
            {
                con.Open();
                SqlCommand cmd = new SqlCommand(
                    "SELECT regNo FROM Registration ORDER BY regNo", con);
                using (SqlDataReader dr = cmd.ExecuteReader())
                {
                    while (dr.Read())
                    {
                        cmbRegNo.Items.Add(dr["regNo"].ToString());
                    }
                }
            }
        }

        private string GetGender()
        {
            return rbMale.Checked ? "Male" : "Female";
        }

        private bool FieldsEmpty()
        {
            return string.IsNullOrWhiteSpace(cmbRegNo.Text)
                || string.IsNullOrWhiteSpace(txtFirstName.Text)
                || string.IsNullOrWhiteSpace(txtLastName.Text)
                || (!rbMale.Checked && !rbFemale.Checked)
```

All rights reserved © ESOFT Metro Campus

```
            || string.IsNullOrWhiteSpace(txtAddress.Text)
            || string.IsNullOrWhiteSpace(txtEmail.Text)
            || string.IsNullOrWhiteSpace(txtMobile.Text)
            || string.IsNullOrWhiteSpace(txtHome.Text)
            || string.IsNullOrWhiteSpace(txtParentName.Text)
            || string.IsNullOrWhiteSpace(txtNIC.Text)
            || string.IsNullOrWhiteSpace(txtContactNo.Text);
    }

    private void AddParameters(SqlCommand cmd)
    {
        cmd.Parameters.AddWithValue("@reg", int.Parse(cmbRegNo.Text));
        cmd.Parameters.AddWithValue("@fn", txtFirstName.Text);
        cmd.Parameters.AddWithValue("@ln", txtLastName.Text);
        cmd.Parameters.AddWithValue("@dob", dtpDOB.Value);
        cmd.Parameters.AddWithValue("@gender", GetGender());
        cmd.Parameters.AddWithValue("@addr", txtAddress.Text);
        cmd.Parameters.AddWithValue("@email", txtEmail.Text);
        cmd.Parameters.AddWithValue("@mob", long.Parse(txtMobile.Text));
        cmd.Parameters.AddWithValue("@home", long.Parse(txtHome.Text));
        cmd.Parameters.AddWithValue("@pn", txtParentName.Text);
        cmd.Parameters.AddWithValue("@nic", txtNIC.Text);
        cmd.Parameters.AddWithValue("@cn", long.Parse(txtContactNo.Text));
    }

    private void ClearFields()
    {
        cmbRegNo.SelectedIndex = -1;
        cmbRegNo.Text = "";
        txtFirstName.Clear();
        txtLastName.Clear();
        dtpDOB.Value = DateTime.Now;
        rbMale.Checked = false;
        rbFemale.Checked = false;
        txtAddress.Clear();
        txtEmail.Clear();
        txtMobile.Clear();
        txtHome.Clear();
        txtParentName.Clear();
        txtNIC.Clear();
        txtContactNo.Clear();
        cmbRegNo.Focus();
    }

    // ---------- REGISTER ----------
    private void btnRegister_Click(object sender, EventArgs e)
    {
        if (FieldsEmpty())
        {
            MessageBox.Show("Fields are empty, please fill them all", "Warning",
                MessageBoxButtons.OK, MessageBoxIcon.Warning);
            return;
        }
        try
        {
            using (SqlConnection con = new SqlConnection(ConnStr))
            {
                con.Open();
                string sql = @"INSERT INTO Registration
                    (regNo, firstName, lastName, dateOfBirth, gender, address, email,
                     mobilePhone, homePhone, parentName, nic, contactNo)
                    VALUES (@reg, @fn, @ln, @dob, @gender, @addr, @email,
                            @mob, @home, @pn, @nic, @cn)";
                SqlCommand cmd = new SqlCommand(sql, con);
                AddParameters(cmd);
```

All rights reserved © ESOFT Metro Campus

```
                cmd.ExecuteNonQuery();
            }
            MessageBox.Show("Record Added Succesfully", "Register Student",
                MessageBoxButtons.OK, MessageBoxIcon.Information);
            LoadRegNumbers();
            ClearFields();
        }
        catch (Exception ex)
        {
            MessageBox.Show(ex.Message, "Error",
                MessageBoxButtons.OK, MessageBoxIcon.Error);
        }
    }

    // ---------- UPDATE ----------
    private void btnUpdate_Click(object sender, EventArgs e)
    {
        if (FieldsEmpty())
        {
            MessageBox.Show("Fields are empty, please fill them all", "Warning",
                MessageBoxButtons.OK, MessageBoxIcon.Warning);
            return;
        }
        try
        {
            int rows;
            using (SqlConnection con = new SqlConnection(ConnStr))
            {
                con.Open();
                string sql = @"UPDATE Registration SET firstName=@fn, lastName=@ln,
                    dateOfBirth=@dob, gender=@gender, address=@addr, email=@email,
                    mobilePhone=@mob, homePhone=@home, parentName=@pn, nic=@nic,
                    contactNo=@cn WHERE regNo=@reg";
                SqlCommand cmd = new SqlCommand(sql, con);
                AddParameters(cmd);
                rows = cmd.ExecuteNonQuery();
            }
            if (rows > 0)
            {
                MessageBox.Show("Record Updated Succesfully", "Update Student",
                    MessageBoxButtons.OK, MessageBoxIcon.Information);
            }
            else
            {
                MessageBox.Show("No record found for this Reg No", "Warning",
                    MessageBoxButtons.OK, MessageBoxIcon.Warning);
            }
        }
        catch (Exception ex)
        {
            MessageBox.Show(ex.Message, "Error",
                MessageBoxButtons.OK, MessageBoxIcon.Error);
        }
    }

    // ---------- DELETE ----------
    private void btnDelete_Click(object sender, EventArgs e)
    {
        if (string.IsNullOrWhiteSpace(cmbRegNo.Text))
        {
            MessageBox.Show("Please select a Reg No to delete", "Warning",
                MessageBoxButtons.OK, MessageBoxIcon.Warning);
            return;
        }
```

All rights reserved © ESOFT Metro Campus

```
            DialogResult result = MessageBox.Show(
                "Are you sure, Do you really want to Delete this Record...?",
                "Delete", MessageBoxButtons.YesNo, MessageBoxIcon.Question);

            if (result != DialogResult.Yes)
            {
                return;
            }
            try
            {
                using (SqlConnection con = new SqlConnection(ConnStr))
                {
                    con.Open();
                    SqlCommand cmd = new SqlCommand(
                        "DELETE FROM Registration WHERE regNo=@reg", con);
                    cmd.Parameters.AddWithValue("@reg", int.Parse(cmbRegNo.Text));
                    cmd.ExecuteNonQuery();
                }
                MessageBox.Show("Record Deleted Succesfully", "Delete Student",
                    MessageBoxButtons.OK, MessageBoxIcon.Information);
                LoadRegNumbers();
                ClearFields();
            }
            catch (Exception ex)
            {
                MessageBox.Show(ex.Message, "Error",
                    MessageBoxButtons.OK, MessageBoxIcon.Error);
            }
        }

        // ---------- SEARCH (when a Reg No is selected) ----------
        private void cmbRegNo_SelectedIndexChanged(
            object sender, EventArgs e)
        {
            if (cmbRegNo.SelectedItem == null)
            {
                return;
            }
            try
            {
                using (SqlConnection con = new SqlConnection(ConnStr))
                {
                    con.Open();
                    SqlCommand cmd = new SqlCommand(
                        "SELECT * FROM Registration WHERE regNo=@reg", con);
                    cmd.Parameters.AddWithValue("@reg",
                        int.Parse(cmbRegNo.SelectedItem.ToString()));
                    using (SqlDataReader dr = cmd.ExecuteReader())
                    {
                        if (dr.Read())
                        {
                            txtFirstName.Text = dr["firstName"].ToString();
                            txtLastName.Text = dr["lastName"].ToString();
                            dtpDOB.Value = Convert.ToDateTime(dr["dateOfBirth"]);
                            if (dr["gender"].ToString() == "Male")
                                rbMale.Checked = true;
                            else
                                rbFemale.Checked = true;
                            txtAddress.Text = dr["address"].ToString();
                            txtEmail.Text = dr["email"].ToString();
                            txtMobile.Text = dr["mobilePhone"].ToString();
                            txtHome.Text = dr["homePhone"].ToString();
                            txtParentName.Text = dr["parentName"].ToString();
                            txtNIC.Text = dr["nic"].ToString();
                            txtContactNo.Text = dr["contactNo"].ToString();
```

All rights reserved © ESOFT Metro Campus

```
                        }
                    }
                }
            }
            catch (Exception ex)
            {
                MessageBox.Show(ex.Message, "Error",
                    MessageBoxButtons.OK, MessageBoxIcon.Error);
            }
        }

        // ---------- CLEAR ----------
        private void btnClear_Click(object sender, EventArgs e)
        {
            ClearFields();
        }

        // ---------- LOGOUT ----------
        private void lnkLogout_LinkClicked(
            object sender, LinkLabelLinkClickedEventArgs e)
        {
            loggingOut = true;
            foreach (Form f in Application.OpenForms)
            {
                if (f is LoginForm login)
                {
                    login.ResetForm();
                    login.Show();
                    break;
                }
            }
            this.Close();
        }

        // ---------- EXIT ----------
        private void lnkExit_LinkClicked(
            object sender, LinkLabelLinkClickedEventArgs e)
        {
            DialogResult result = MessageBox.Show(
                "Are sure, Do you really want to Exit...?", "Exit",
                MessageBoxButtons.YesNo, MessageBoxIcon.Question);

            if (result == DialogResult.Yes)
            {
                Application.Exit();
            }
        }
    }
}
```

All rights reserved © ESOFT Metro Campus

7. Message Boxes

The table below lists every message box in the system. Take a screenshot of each one while the program is running and insert it in the last column.

Message box	Shown when	Title and text	Screenshot
Error message	Wrong username and/or password	Invalid login Details – Invalid Login credentials, please check Username and Password and try again	[Insert]
Exit message	Exit button or Exit link is clicked	Exit – Are sure, Do you really want to Exit...?	[Insert]
Register message	Student registered successfully	Register Student – Record Added Succesfully	[Insert]
Update message	Record updated successfully	Update Student – Record Updated Succesfully	[Insert]
Delete confirmation	Delete button is clicked	Delete – Are you sure, Do you really want to Delete this Record...?	[Insert]
Delete completed	User clicks Yes on delete confirmation	Delete Student – Record Deleted Succesfully	[Insert]
Empty fields warning	Register or Update with empty fields	Warning – Fields are empty, please fill them all	[Insert]

All rights reserved © ESOFT Metro Campus

8. Testing

The software was tested using the test cases below. Fill in the actual result and the status after you run each test yourself.

No	Test case	Expected result	Actual result	Status
1	Run the project	Login form appears first	[ ]	[ ]
2	Login with Admin / Campusone@123	Login form hides; Registration form opens	[ ]	[ ]
3	Login with a wrong password	Error message box appears	[ ]	[ ]
4	Login with a wrong username	Error message box appears	[ ]	[ ]
5	Click Clear on the Login form	Both boxes cleared; cursor in username box	[ ]	[ ]
6	Click Exit, then No	Dialog closes; system stays open	[ ]	[ ]
7	Click Exit, then Yes	System closes	[ ]	[ ]
8	Fill all fields and click Register	Record saved; success message	[ ]	[ ]
9	Click Register with empty fields	Warning message appears	[ ]	[ ]
10	Open the Reg No list and select a number	All other fields are filled from the database	[ ]	[ ]
11	Change a value and click Update	Record updated; update message	[ ]	[ ]
12	Click Delete, then No	Record is not deleted	[ ]	[ ]
13	Click Delete, then Yes	Record deleted; delete message	[ ]	[ ]
14	Click Clear on the Registration form	All fields emptied	[ ]	[ ]
15	Click Logout	Registration form closes; Login form appears	[ ]	[ ]

Screenshots of the running system

All rights reserved © ESOFT Metro Campus

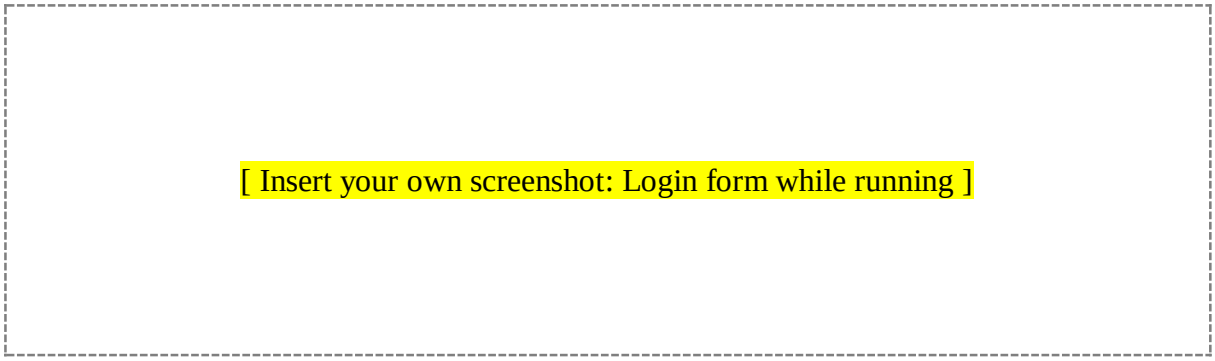


Figure 8.1 – Login form (running)

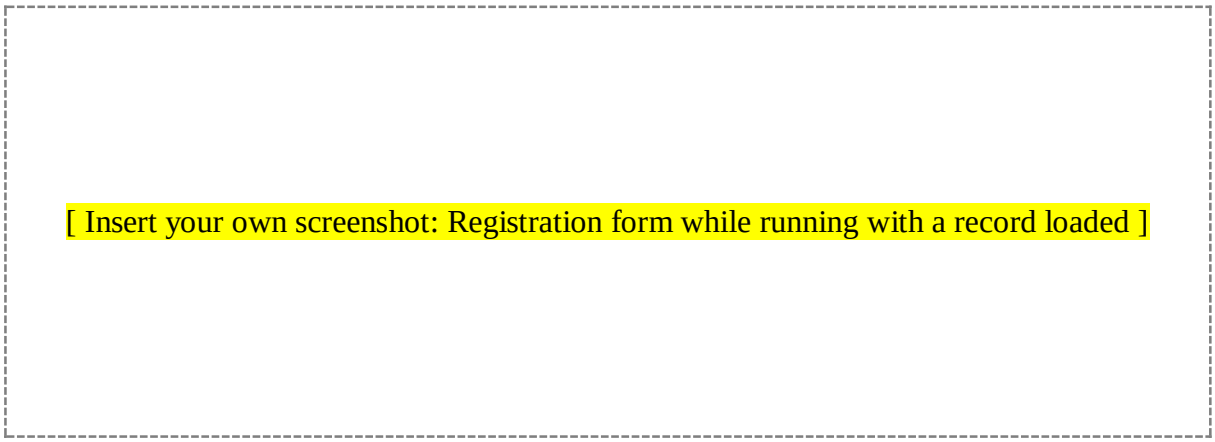


Figure 8.2 – Registration form (running)

All rights reserved © ESOFT Metro Campus

9. Conclusion

[Write 1–2 paragraphs in your own words: what the system does, what you learned while building it, and any extra features or improvements you added.]

10. References

[If you used any outside sources (websites, books, tutorials), list them here in the Harvard referencing style and cite them in your text. If you used none, delete this section.]

All rights reserved © ESOFT Metro Campus